

Henrique dos Santos Teixeira

henrique.dsteixeira.dev@gmail.com | www.henriquesantosteixeira.dev | github.com/ushouldrun | LinkedIn

EXPERIENCE

- Researcher - AIMaCoV: ONNX Graph Mapping to RISC-V CGRA** *INESC TEC* 02/2026 – now | Porto, Portugal
- Proactively advocated to lead a new research initiative, mentoring 2 undergraduate interns on pipeline management and algorithmic strategy.
 - Guided the design of a GNN-guided DFG partitioning flow for mapping ONNX computation graphs onto a RISC-V spatial CGRA, learning optimal partitions from SAT-generated ground truth.
 - Mentored work achieved 33% edge-cut reduction over the unguided baseline, 98% mapping success vs. 15% for greedy list scheduling and 50× speedup over exact SAT (~0.25 s/graph).
- Internship - LNS on a RISC-V SoC via High Level Synthesis** *INESC TEC* 02/2026 – 06/2026 | Porto, Portugal
- Extended a prior LNSU implementation by adding comparison operations and FP/BF ↔ LNS conversion functions, approximated via greedy-generated linear splines with analytically derived error bounds.
 - Built a header-only C++ LNS emulation library (lnssim.hpp) with XF and XMB spline backends, supporting lns8 Q4.3 and lns16 Q8.7 and a matching BF emulator (bfloatsim.hpp) for direct benchmarking.
 - Validated lns16 Q8.7 for Llama 2 inference: direct fp32 → lns16 weight conversion without retraining produced coherent text on TinyStories 15M and 42M, with hybrid fp32 accumulation isolating LNS to matrix multiplications.
- Researcher - AIQ-Ready: Accelerator for Compute-Heavy Nodes in ONNX Models** *INESC TEC* 09/2025 – 01/2026 | Porto, Portugal
- Assessed viability of converting ONNX models to C via ONNX2C and profiled compute-heavy nodes in the resulting output.
 - Selected bottleneck nodes for hardware acceleration on XHeep/RISC-V and evaluated combined SW + accelerated results.
- Internship - AI Acceleration on a RISC-V SoC via High Level Synthesis** *INESC TEC* 06/2025 – 07/2025 | Porto, Portugal
- Best CTM Internship – Computer Science and Engineering winner.
 - Deployed a Logarithmic Number System Unit to a RISC-V CPU via HLS in C++, with addition/subtraction approximations using only 192 bytes of lookup tables combined.
 - Integrated into the RISC-V pipeline and tested on FPGA at 125 MHz.
- Web Application Developer** *Tuna de Ciências do Porto* 02/2025 – now | Porto, Portugal
- Co-developed and maintain a full-stack web application to manage events, members, instruments and organizational workflow for a student musical group, automating previously manual workflows.
 - Informally act as technical lead, managing a team of 8 student contributors, handling onboarding, development workflow and self-hosting infrastructure with Cloudflare Tunnel and JWT authentication.
 - Built with React (Vite), Go and MySQL, containerized with Docker Compose.
- Academic Internship - Lambda Calculus Interpreter** *FCUP* 02/2025 – 06/2025 | Porto, Portugal
- Implemented a C interpreter for Lambda Calculus compiled to SK Combinators, combining both theories to build translating and reducing functions over a custom program tree.

PAPERS

- GNN-Guided Graph Partitioning for Spatial CGRA Mapping** *INForum 2026 (CPDLA)* 09/2026
- Co-author (with D. Lopes Günes, G. Vareiro de Oliveira, N. Paulino); 33% edge-cut reduction, 98% mapping success, 50× SAT speedup.

PROJECTS

Systems & Infrastructure

- Lightweight Cloud Job Platform on GCP**
- CLI-driven platform on GCP submitting shell commands as jobs on auto-provisioned GCE worker VMs, with a FastAPI backend, Redis queue, autoscaler daemon (scale-to-zero), JWT auth, rate limiting and IAP tunneling.
- LNS Library**
- C++ library implementing 8-bit and 16-bit Logarithmic Number System arithmetic, with exact multiply/divide via exponent addition, spline-based LUT approximation for add/sub and both a hardware compiler header (custom RISC-V instructions) and a software simulation API for benchmarking against float32 and bfloat16.
- RISC-V Assembler with Custom Instructions**
- Assembler for RISC-V supporting 32IM instructions and custom LNSU instructions from INESC TEC internship.
- Kotlin Compiler**
- Compiler for a subset of Kotlin (statements, expressions, enums, functions) targeting MIPS assembly.
- Memory Allocator Library**
- Custom C library implementing an Arena (O(1) linear) and a Pool (best-fit, O(F)) allocator with coalescing, realloc and strdup support; used in the Kotlin Compiler and SK Interpreter.

ML & Data Science

- Autonomous Driving: DQN vs PPO**
- Trained DQN and PPO agents for autonomous lane-following and obstacle avoidance in Webots using a custom multi-branch CNN (LaneCNN) fusing Camera and LiDAR inputs; PPO achieved 60% success on unseen tracks vs. 0% for DQN. Python, PyTorch.
- UrbanSound8K Audio Classification**
- Built MLP and CNN classifiers for 10-class urban sound classification using the UrbanSound8K dataset; evaluated via 10-fold cross-validation with librosa-based feature extraction. Python, PyTorch.
- Lung Cancer Classification**
- Developed deep learning classifiers for lung cancer detection from CT scan data. Python, PyTorch.
- Robust Naive Bayes for Multiclass Classification**
- Modified and benchmarked a Naive Bayes classifier from scratch (no scikit-learn) to improve robustness to multiclass imbalance; compared against standard implementation across multiple datasets. Python, NumPy.
- Connect Four: MCTS and Decision Tree AI**
- Implemented MCTS with UCT and a Decision Tree (ID3) trained on MCTS self-play data, with a bitboard game state representation. Python, NumPy.

EDUCATION

- Bachelor's Degree in Artificial Intelligence and Data Science** *FCUP* 09/2023 – 07/2026 | Porto, Portugal
- 2× winner of “Bolsas PRR «Impulso Jovens STEAM» - Distinção Académica” scholarship.
 - Relevant Courses: **Compilers** (Extra.), **AI**, **Machine Learning I & II**, **Data Structures**, **Algorithms**

TOOLS

Systems/Hardware: RISC-V, HLS, FPGAs, Docker
Infrastructure/Cloud: GCP, Redis, FastAPI, Cloudflare Tunnels, JWT
ML/Data: PyTorch, Pandas, NumPy, librosa

PROGRAMMING LANGUAGES

C | C++ | Python | Go | JavaScript